EXP.NO: 7

AIM:

To create a Servlet-based Personal Portfolio Builder application that stores user preferences (such as theme and language) using cookies and retrieves them on subsequent visits.

ALGORITHM:

1. Start the application.
2. Accept user preferences (theme/language) through a form.
3. Create cookies for each preference.
4. Set cookie expiry time.
5. Store cookies in the client browser.
6. When the user revisits:
   • Read cookies from request.
   • Check stored values.
7. If cookies exist:
   • Display stored preferences.
8. If cookies not found:
   • Show default values.
9. Stop the application.

STEPS:

1. Design a UI to select Theme and Language.
2. Create Servlet to handle form submission.
3. Store preferences using cookies.
4. Set expiry time for cookies.
5. Send cookies to browser.
6. Retrieve cookies on next visit.
7. Display preferences dynamically.
8. Show default values if no cookies exist.

Program

---

index.html (Preference Page)

HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>Portfolio Preferences</title>
    <style>
        body {
            font-family: Arial;
            background: #dbe2ef;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .card {
            background: white;
            padding: 30px;
            border-radius: 10px;
            width: 320px;
            text-align: center;
            box-shadow: 0 0 10px rgba(0,0,0,0.2);
        }

        select, button {
            width: 100%;
            padding: 10px;
            margin: 10px 0;
        }

        button {
            background: #3f72af;
            color: white;
            border: none;
            border-radius: 5px;
        }

        a {
```

```
            display: block;
            margin-top: 10px;
            color: blue;
        }
    </style>
</head>

<body>

<div class="card">
    <h2>Portfolio Builder</h2>

    <form action="PreferenceServlet" method="post">
        <label>Select Theme</label>
        <select name="theme">
            <option>Light</option>
            <option>Dark</option>
        </select>

        <label>Select Language</label>
        <select name="language">
            <option>English</option>
            <option>Tamil</option>
        </select>

        <button type="submit">Save Preferences</button>
    </form>

    <a href="DisplayServlet">View Portfolio</a>
</div>

</body>
</html>
```

PreferenceServlet.java

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String theme = request.getParameter("theme");
        String language = request.getParameter("language");

        Cookie themeCookie = new Cookie("theme", theme);
        Cookie langCookie = new Cookie("language", language);

        themeCookie.setMaxAge(60 * 60 * 24);
        langCookie.setMaxAge(60 * 60 * 24);

        response.addCookie(themeCookie);
        response.addCookie(langCookie);

        PrintWriter out = response.getWriter();
        out.println("<h3>Preferences Saved!</h3>");
        out.println("<a href='DisplayServlet'>Go to Portfolio</a>");
    }
}
```

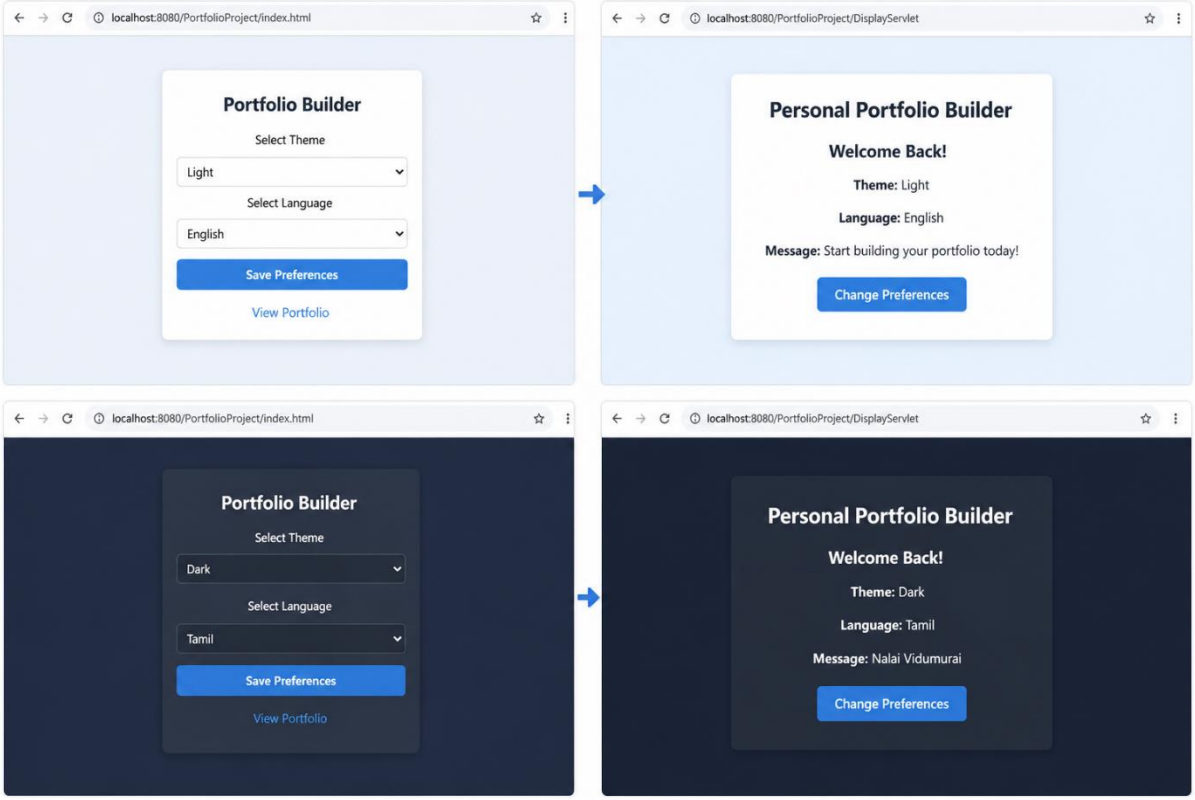
DisplayServlet.java

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class DisplayServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String theme = "Light";
        String language = "English";

        Cookie[] cookies = request.getCookies();

        if (cookies != null) {
```

```
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) {
                    theme = c.getValue();
                }
                if (c.getName().equals("language")) {
                    language = c.getValue();
                }
            }
        }

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String bg = theme.equals("Dark") ? "#1e1e2f" : "#dbe2ef";
        String card = theme.equals("Dark") ? "#2c2c3c" : "#ffffff";
        String text = theme.equals("Dark") ? "#ffffff" : "#000000";

        out.println("<html><body style='background:" + bg + "; text-align:center; font-
family:Arial;'>");

        out.println("<div style='margin:60px auto; padding:30px; width:320px;
background:" + card + "; border-radius:10px;'>");

        out.println("<h2>Personal Portfolio Builder</h2>");
        out.println("<h3>Welcome Back!</h3>");
        out.println("<p><b>Theme:</b> " + theme + "</p>");
        out.println("<p><b>Language:</b> " + language + "</p>");

        if (language.equals("Tamil")) {
            out.println("<p><b>Message:</b> Nalai Vidumurai</p>");
        } else {
            out.println("<p><b>Message:</b> Start building your portfolio today!</p>");
        }

        out.println("<a href='index.html' style='display:inline-block; margin-top:15px;
padding:10px; background:#3f72af; color:white; text-decoration:none;'>Change
Preferences</a>");

        out.println("</div></body></html>");
    }
}
```

OUTPUT:



RESULT:

The Personal Portfolio Builder application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.